

Project 1



From CO₂ to Insight

A Data-Driven Exploration of CO₂ Electroreduction

Bright Jaato

October 2, 2026

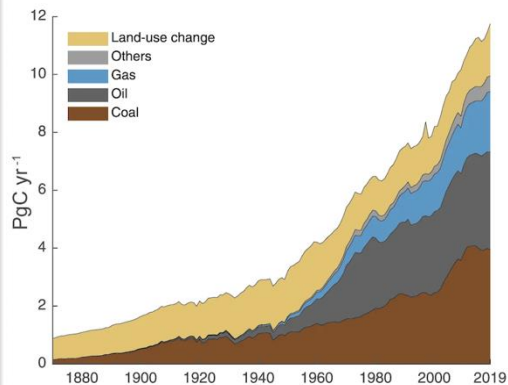
Data Science & Machine Learning

Presentation Overview

- Introduction
- Dataset & Research Questions
- From Raw Data to SQL Analysis
- Key Findings
- Conclusions
- Limitations and Next Steps

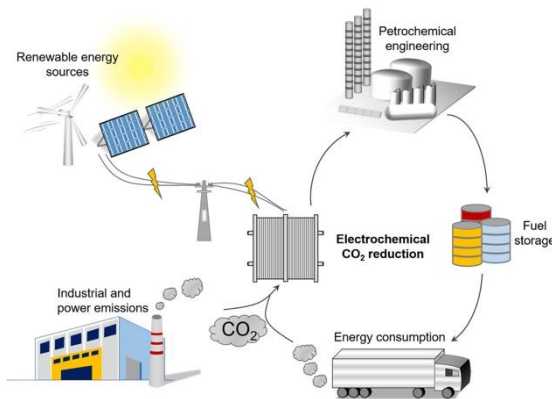
From CO₂ Challenge to Data Challenge

CO₂ Challenge



Global anthropogenic CO₂ emissions

CO₂RR Opportunity



CO₂ + renewable electricity → chemicals and fuels

Data Challenge

- Many catalysts and catalyst combinations
- Different products and selectivities
- Different electrolytes and reactors
- Operating conditions vary across studies
- Metrics are reported inconsistently

→ **Direct comparison is difficult**

Can we structure this literature data to identify promising CO₂RR systems?

Data & Research Questions

Dataset

Low-temperature dataset

101 records

Catalyst • Product • FE • Current density • Cost

Second experimental dataset

181 records

Catalyst • Electrolyte • Temperature • Total
current density

Questions

RQ1: How does performance vary across catalyst–product combinations?

RQ2: Which combinations have high FE, high current density, and lower reported cost?

RQ3: How does current density vary across catalyst–electrolyte systems?

RQ4 How do systems and current density differ across temperature groups?

From Raw Data to SQL Analysis

How raw literature data became a structured database for analysis

Explore



Prepare



Build Database



Analyse

Understand the raw data and identify issues.

- Inspect data structure
- Check missing values and duplicates

Clean, standardise, and transform the data.

- Standardise names and formats
- Preserve meaningful missing values
- Create clean, database-ready data

Create related tables, keys, and SQLite database.

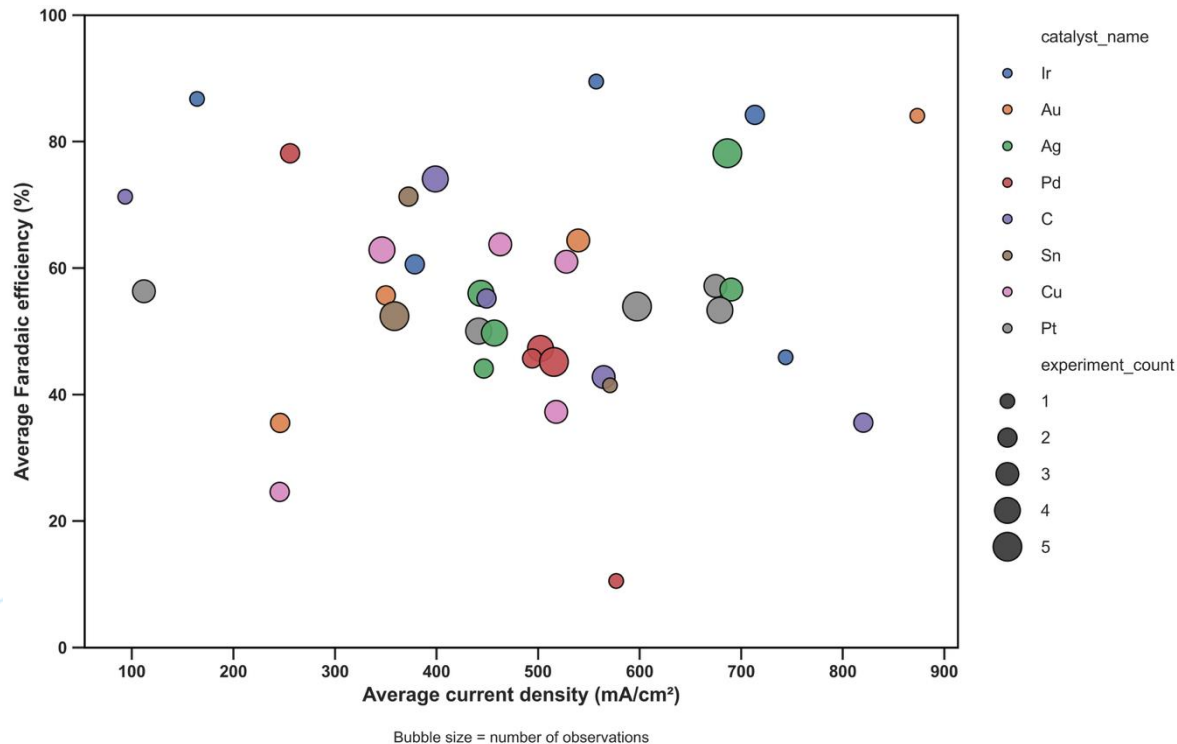
- Create 5 related tables
- Define primary and foreign keys
- Build and verify in SQLite

Use SQL and visualisation to answer the research questions.

- Query the database
- Compare systems
- Turn results into visual insights

Raw data → prepared data → relational database → SQL insights

Finding 1: Performance Depends on More Than One Metric

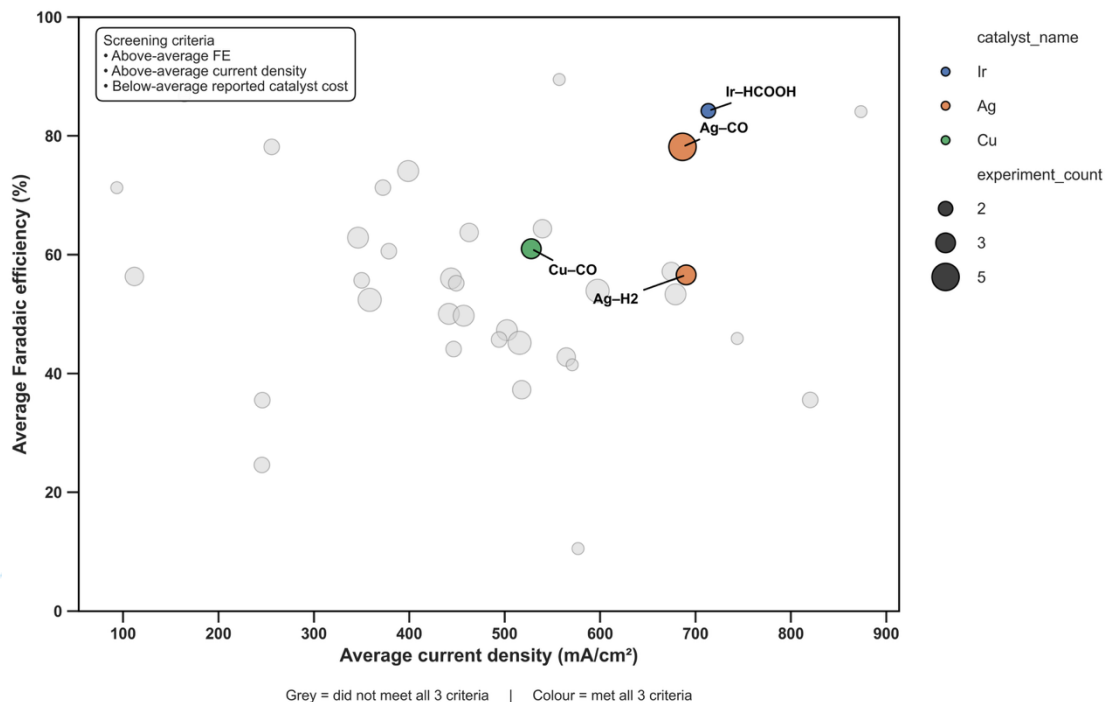


- FE and current density vary widely
- High FE does not always mean high current density
- Some averages are based on few observations

No single metric is enough to identify the best system.

Catalyst-product combination = catalyst + product produced (eg. Ag-CO)

Key Finding 2: Screening Narrows the Choices



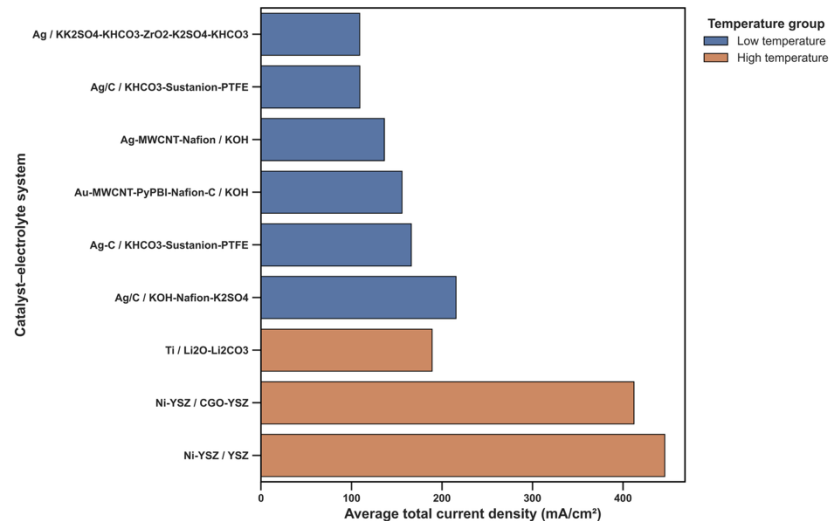
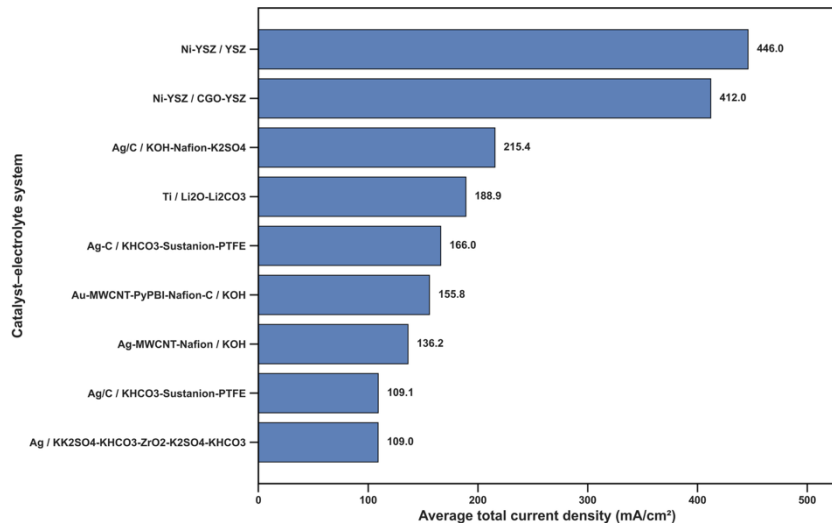
- Only 4 combinations met all three criteria
- Ir-HCOOH had the highest FE among the four, with high current density
- Ag-CO combined better performance with more supporting observations

Performance and cost screening reduced the choices to 4 systems.

HCOOH = Formic acid
CO = Carbon monoxide

Finding 3: The Whole System Matters

How do Catalyst-electrolyte performance are relate to current density Systems by temperature group



- Ni-YSZ/YSZ and Ni-YSZ/CGO-YSZ had the highest average current densities
- The same catalyst showed different performance with different electrolytes

Catalyst, electrolyte, and operating conditions must be considered together

Conclusions

- Performance must be judged using several metrics
- Only a few systems remained promising after performance and cost screening
- Catalyst, electrolyte, and operating conditions must be interpreted together

Data-driven screening can turn a large CO₂RR dataset into a smaller set of promising systems for further study.

Limitations and Next Steps

Limitations

- The studies used different experimental conditions
- Some groups have only a few data points
- Voltage and FE were not reported together in the second dataset

Next Steps

- Add more studies with similar experimental conditions
- Compare full data ranges, not only averages
- Explore simple statistical and machine-learning models

More consistent data will allow better comparisons and more reliable predictions.

Thank you

Questions?



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From complex CO₂RR literature to structured, data-driven insight

